

Manufacturing

The Challenge

Manufacturing wireless components and devices (e.g., cell phones) and BTS chipsets can be a daunting task. An especially critical challenge is verifying product performance prior to delivery to the customer. The increasing complexity of today's emerging wireless communications standards means that more testing is required and that testing often requires more time. Test time is directly related to cost—something that manufacturers are continually looking to reduce. Running tests using a fast spectrum analyzer is one way to cut test time and in turn, cost. Test accuracy is also critical since it allows engineers to optimize their test budgets for manufacturing variances rather than test instrument uncertainties, thereby increasing yield, maximizing productivity and minimizing cost. Overcoming these test challenges requires use of an economically-priced signal analysis solution with exceptional speed and measurement accuracy.

The Solution

As a high-speed, high-performance, cost-effective solution, the MS2830A Series Spectrum Analyzer/Signal Analyzer is well suited to addressing the challenges facing today's manufacturer. It captures wideband signals and, with its FFT technology, supports multifunction signal analysis in both the time and frequency domains. Behavior in the time domain that cannot be handled by a sweep type spectrum analyzer can be checked in the frequency domain. A wide frequency can be analyzed using sweep type spectrum analysis functions, while detailed signal analysis of a specific frequency band is also supported. Moreover, a built-in signal generator function outputs both continuous wave (CW) and modulated signals for use as a reference signal source for evaluating Rx characteristics.

The MS2830A-040/041/043 is well suited for Tx test, offering the excellent dynamic range performance necessary for spurious test and a wide range of modulation analysis software (e.g., LTE FDD/TDD, WiMAX, GSM/EDGE, and W-CDMA/HSPA/HSPA Evolution). The MS2830A-044/045 is best suited for the development and manufacture of microwave products, featuring both spectrum and signal analysis capabilities.

The MS2830A-044/045 Signal Analyzer includes a spectrum analyzer function for measuring up to 110 GHz using an external mixer based on the 26.5 GHz/43 GHz upper frequency limit. Installing the 10 MHz/31.25 MHz bandwidth analysis option adds signal analyzer functions for checking phenomena that are hard to check using a spectrum analyzer, such as frequency vs. time, phase vs. time, spectrogram, and CCDF. Optional measurement software supports modulation analysis. The MS2830A-044/045 can be customized to support a range of application-specific measurements, such as measurement of weaker signals and catching the instant change of the frequency/phase/amplitude/spectrum of microwave signals.

For more information go to MS2830A Signal Analyzer product page.



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